SMART TV SALES ANALYSIS

Power BI Student Project Report

Student Name	Nirmal Singh Chauhan
Registration Number	PIET24CD038
Project Title	Smart TV Sales Analysis using Microsoft Power BI
Analysis Scope	Brand, Screen Size, Price, Rating and Sales
Tools	Microsoft Excel, Power BI Desktop and DAX

Project Overview

This project analyses Smart TV sales data in five required areas: brand, screen size, price, rating and sales. Power BI is used to prepare measures, compare products and present results in a single-page dashboard. The dataset is synthetic practice data and does not represent actual retailer transactions.

1. Dataset Description

The dataset contains 10,000 synthetic sales records. Each row is a sales record, and SalesRevenueINR is calculated as PriceINR multiplied by UnitsSold.

Column	Description	Use
SaleID	Unique sales record ID	Count records
Brand	TV brand	Brand comparison
ScreenSizeInches	Screen size in inches	Size-wise sales
PriceINR	TV price in rupees	Price analysis
Rating	Product rating from 3.0 to 5.0	Average rating
UnitsSold	Units in a sales record	Sales volume
SalesRevenueINR	Price × units sold	Revenue analysis

2. Project Objectives

• Compare revenue across TV brands.

• Identify screen sizes contributing the most sales revenue.

• Explore price, units sold and revenue relationships.

• Compare average ratings by brand.

• Compare units sold by brand.

• Build a clear Power BI dashboard.

3. Tools and Technologies

Excel: data storage. **Power BI Desktop:** data import, modelling and visualisation. **DAX:** summary measures.

4. Data Preparation and Cleaning

Verify data types in Power Query: Brand as Text, Rating as Decimal Number, and SaleID, ScreenSizeInches, PriceINR, UnitsSold and SalesRevenueINR as numeric. Check blanks, errors, duplicate IDs and confirm that SalesRevenueINR equals PriceINR multiplied by UnitsSold.

Note: These are validation recommendations; only claim a problem was removed if you actually checked and corrected it.

5. Data Model

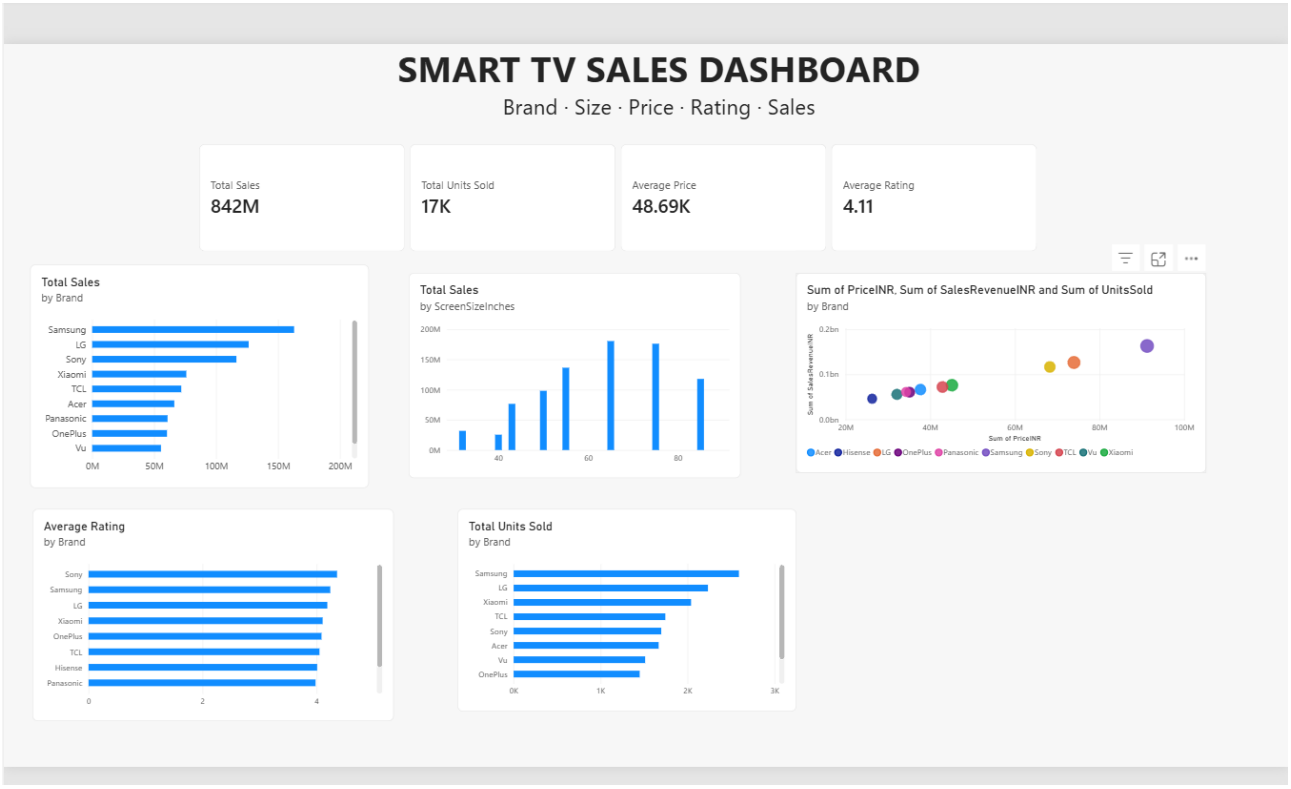
The project uses one table named SmartTVSales, so no relationships are required. SaleID identifies each record; Brand, ScreenSizeInches, PriceINR and Rating support grouping and analysis.

6. DAX Measures

Measure	DAX Formula
Total Sales	`Total Sales = SUM(SmartTVSales[SalesRevenueINR])`
Total Units Sold	`Total Units Sold = SUM(SmartTVSales[UnitsSold])`
Average Price	`Average Price = AVERAGE(SmartTVSales[PriceINR])`
Average Rating	`Average Rating = AVERAGE(SmartTVSales[Rating])`
Total Records	`Total Records = DISTINCTCOUNT(SmartTVSales[SaleID])`
Average Screen Size	`Average Screen Size = AVERAGE(SmartTVSales[ScreenSizeInches])`

7. Dashboard Design

The dashboard includes KPI cards for total sales, units sold, average price and average rating, plus visuals for brand revenue, screen-size revenue, price versus revenue, average rating by brand and units sold by brand.



8. Visuals and Field Mapping

Visual	Fields	Purpose
KPI cards	[Total Sales], [Total Units Sold], [Average Price], [Average Rating]	Overall summary
Sales by Brand	Brand + [Total Sales]	Compare revenue by brand
Sales by Size	ScreenSizeInches + [Total Sales]	Compare revenue by TV size
Price vs Revenue	PriceINR, SalesRevenueINR, UnitsSold; Brand legend	Explore price/revenue and volume
Average Rating	Brand + [Average Rating]	Compare ratings
Units Sold	Brand + [Total Units Sold]	Compare sales volume

9. Key Findings

The following values are displayed in the supplied dashboard screenshot. They describe this synthetic dataset, not actual market performance.

Metric	Displayed value	Interpretation
Total Sales	842M	Approximately ■842 million total sales revenue displayed.
Total Units Sold	17K	Approximately 17,000 units sold displayed.
Average Price	48.69K	Average price displayed is approximately ■48,690.
Average Rating	4.11	Average rating displayed is 4.11 out of 5.
Brand revenue	Samsung leads visible chart	Samsung has the longest bar in the supplied chart.
Screen size	55-inch and 75-inch bars are tall	These sizes show high revenue in the visual; verify exact ranking from the data.
Rating	Sony leads visible chart	Sony appears to have the highest average rating in the screenshot.

10. Conclusion

This project demonstrates how Power BI can summarise and compare Smart TV sales across brands and screen sizes. KPI cards provide an overview, bar and column charts compare revenue and units sold, the rating chart compares average ratings, and the scatter chart explores price and revenue. Together, these visuals cover brand, size, price, rating and sales.

Before final submission, format monetary values in INR, confirm all aggregations, and ensure the scatter chart uses meaningful fields. This dataset has no date column, so time-based sales trends are outside the scope.

11. Suggested Submission Folder Structure

```
Smart-TV-Sales-Project/
■■■ README.md
■■■ Dataset/Smart_TV_Sales_Dataset_10000_Rows.xlsx
■■■ PowerBI/Smart_TV_Sales.pbix
■■■ Screenshots/dashboard.png
■■■ Report/Project_Report.pdf
```

Prepared by Nirmal Singh Chauhan | PIET24CD038